of the disk (analogous to the wind that blows off the Sun's surface) may create a bubble of hot gas inside the cool cloud. The hot gas may then punch orifices in the cool cloud's upper and lower faces and flow out of them. Just as a nozzle on a garden hose collimates outflowing water to form a fast, thin stream, so the orifices in the cool cloud should collimate the outflowing hot gas to form thin jets. The directions of the jets will depend on the locations of the orifices. The most likely locations, if the cool cloud spins about the same axis as the black hole, are along the common spin axis, that is, perpendicular to the plane of the inner part of the accretion disk —and the orifices at these locations will produce jets whose direction is anchored to the black hole's gyroscopic spin.

Second, because the disk is so hot, its internal pressure is very high, and this pressure might puff the disk up until it becomes very thick (Figure 9.7b). In this case, Lynden-Bell pointed out, the orbital motion of the disk's gas will produce centrifugal forces that create whirlpool-like funnels in the top and bottom faces of the disk. These funnels are precisely analogous to the vortex that sometimes forms when water swirls down the drainhole of a bathtub. The black hole is like the drainhole, and the disk's gas is like the water. The faces of the vortex-like funnels should be so hot, because of friction in the gas, that they blow a strong wind off themselves, and the funnels might then collimate this wind into jets, Lynden-Bell reasoned. The jets' directions will be the same as the funnels', which in turn are firmly anchored to the hole's gyroscopic spin axis.

Third, Blandford realized, magnetic field lines anchored in the disk and sticking out of it will be forced, by the disk's orbital motion, to spin around and around (Figure 9.7c). The spinning field lines will assume an outward and upward (or outward and downward) spiraling shape. Electrical forces should anchor hot gas (plasma) onto the spinning field lines; the plasma can slide along the field lines but not across them. As the field lines spin, centrifugal forces should fling the plasma outward along them to form two magnetized jets, one shooting outward and upward, the other outward and downward. Again the jets' directions will be firmly anchored to the hole's spin.